

Programme	BSc Mathematics Honours			
Course Code	MAT2VN102			
Course Title	R PROGRAMMING			
Type of Course	Vocational Minor – Introduction to Data Science			
Semester	II			
Academic Level	100-199			
Course Details	Credit	Lecture/Tutorial	Practical	Total Hours
		per week	per week	
	4	3	2	75
Pre-requisites	Foundations in Mathematics, Programming Fundamentals			
Course Summary	Course aims to provide R programming fundamentals and algorithm writing			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basic programming structure of R, visualization of models and their inference.	U	P	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
CO2	Apply statistical functions, models and their Inferences	Ap	P	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
CO3	Design data model, visualization and inference of dataset to gain insights	C	P	Internal exam/ Assignment/ Seminar/ External/ Practical Assessment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Ext. Marks (70)
I		Introduction to R	10	Min.10
	1	Introduction to R: R Studio, Basic components in R Studio.		
	2	Basic R syntax: variables, data types, operators		
	3	Working with Data structures Vectors, List, Matrices & Arrays, Factors and Data frame		
	4	Control structures (if-else statements, Loops) & Functions		
	5	Measures of Central Tendency & Dispersion		
II		Data Manipulation and Visualization with R	13	Min.20
	6	Importing and exporting data in R (CSV, Excel, Xml, Json, databases)		
	7	Data Cleaning: Exploring raw data, Missing values, Zeros and NAs – Separating, Uniting Columns, String Manipulation, Filling Missing values		
	8	Data manipulation with dplyr: filtering, selecting, mutating, summarizing		
	9	Basic Charts: Pie, Bar, Histogram, Boxplot and Scatterplot		
	10	Data visualization with ggplot2: creating plots (scatter plots, bar plots, line plots)		
	11	Customizing plots and Introduction to other Visualization Packages (ggplot2 extensions, plotly)		
III		Statistical Analysis with R	9	Min.15
	12	Overview of statistical analysis in R		
	13	Descriptive statistics: mean, median, standard deviation, variance		
	14	Probability distributions and random variables		
	15	Hypothesis testing: t-tests, chi-square tests, ANOVA		
	16	Linear regression analysis: simple and multiple regression		

	17	Introduction to statistical modelling with R		
IV		Introduction to Machine Learning with R	13	Min.15
	18	Introduction to machine learning concepts and algorithms		
	19	Supervised learning techniques: classification and regression		
	20	Unsupervised learning techniques: clustering and dimensionality reduction		
	21	Case study – Explore Diamond dataset for prize prediction		
	22	Applied Analytics – HR, Finance & Marketing, Case studies		
		Practical's	30	
	1	Write a R program to take input from user (name, age, occupation, salary) and display the values with datatypes. Also print version of R installation.		
	2	Write a R program to calculate the sum of numbers from 1 to 10.		
	3	Write a R Program to create a list containing a vector, a matrix and a list and write a code for the following. 1) Give names to the elements in the list 2) Add element at the end of the list 3) Remove the second element		

	4	R program to create a data frame of student with four given vectors and write a code 1) to get the structure of a given data frame. 2) to get the statistical summary and nature of the data of a given data frame. 3) to extract specific column from a data frame using column name. 4) to extract first two rows from a given data frame. 5) to extract 3rd and 5th rows with 1st and 3rd columns from a given data frame. 6) to add a new column in a given data frame. 7) to add new row(s) to an existing data frame. 8) to drop column(s) by name from a given data frame. 9) to drop row(s) by number from a given data frame. a) 10) to extract the records whose grade is greater than 9
	5	Write a R program to find biggest of 3 number (if -else)
	6	Write a R program to find sum of elements of vector and to find minimum and maximum elements of vector (loop)
	7	Write a R program to Import a CSV file named 'data.csv' into a data frame named 'data_df'. a) Display the structure of the 'data_df' data frame using the 'str()' function. b) Print the first few rows of the data frame to inspect the data using the 'head()' function. c) Calculate summary statistics (mean, median, min, max) for numerical variables in the data frame using the 'summary()' function.

	8	Write a Program in R for Missing value imputation 1) Load the 'iris' dataset into a data frame named 'iris_df'. 2) Introduce missing values into the 'iris_df' dataset by randomly replacing a certain percentage of values with NA. 3) Display the summary of missing values in the dataset using the 'is.na()' and 'colSums()' functions. 4) Impute missing values in the dataset using a simple technique (e.g., replacing missing values with the mean or median of the corresponding column). 5) Verify that there are no missing values remaining in the dataset after imputation. 6) Compare summary statistics (mean, median, min, max) of the dataset before and after missing value imputation.
	9	Import a dataset from a CSV file and use dplyr to filter rows based on a condition.
	10	Write a R Program to print data in different graph formats (Histogram, Pie, Bar, Boxplot, Scatterplot)
	11	Write a R program to visualize different plot using ggplot 1) Load the 'iris' dataset into a data frame named 'iris_df'. 2) Create a scatter plot of 'Sepal.Length' against 'Sepal.Width' with points colored by 'Species'. 3) Generate a box plot of 'Petal.Length' for each 'Species'. 4) Create a histogram of 'Sepal.Length' with customized bin widths and colors. 5) Generate a density plot of 'Petal.Width' for each 'Species' overlaid on the same plot. 6) Create a bar plot showing the count of each 'Species' in the dataset. 7) Generate a violin plot of 'Petal.Length' for each 'Species' with custom fill colors. 8) Create a line plot showing the trend of 'Sepal.Length' over 'Petal.Length' for each 'Species'. 9) Combine multiple plots into a single visualization using facets based on 'Species'. 10) Customize the appearance of the plots by adding titles, axis labels, legends, and adjusting plot aesthetics (e.g., colors, transparency).
	12	Write a Program to find mean, median, standard deviation and variance

	13	The heights of 6 randomly chosen sailors are 63,65,68,69,71,72 inches. Those of 10 randomly chosen soldiers are 61,62,65,66,69,69,70,71,72,73 inches. Discuss whether this data gives a suggestion that the sailors are taller than soldiers. Aim: To test the claim that sailors are taller than soldiers (t-test)		
	14	Write a R Program to Apply Simple Linear Regression and Multiple Linear Regression		
	15	Write a R Program to Apply K-means clustering algorithm to the data and visualize the clusters.		
		References		
	1	Hands-On Programming with R by Garrett Grolemund		
	2	R Cookbook by Winston Chang, Paul Teetor, and Joseph Adler		
	3	Beginning R: The Statistical Programming Language by Mark Gardener		
	4	The Art of R Programming by Norman Matloff		
	5	Advanced R by Hadley Wickham		

Note: Proofs of all the results are exempted for the end semester exam.

Mapping of COs with PSOs and POs :

	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	3	3	2	3	3	3	2	2
CO 2	3	3	3	2	3	3	3	2	2
CO 3	3	3	3	2	3	3	3	2	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓